



University of
Applied Sciences

DDM2ILV - Data and Decision Making

**ECTS: 6 | Contact hours per week: 4 |
Semester: 2 | SS26**

Lecturer(s) / Lehrende(r)

Dr. Karl Michael HÖFERL

Learning outcomes

At the end of this module students are able to:

- use spreadsheet programs and statistical software to solve economic problems using formulas, functions and diagrams,
- describe and analyse the distribution of a variable, identify & interpret relationships among variables,
- explain probability concepts and distributions, make decisions under conditions of uncertainty,
- carry out random samples and derive conclusions using statistical hypothesis tests.

Course description

- Data Analysis
- Probability & Decision Making under uncertainty
- Statistical Inference
- Regression Analysis & Time Series Forecasting

Teaching concept

Teaching Concept:

In the module, students are introduced to the defined learning objectives through various teaching methods. The combination of lectures, discussions, exercises, and self-study creates a holistic learning experience aimed at thoroughly preparing students for the analysis of economic problems and data-driven decision-making.

Lecture with discussion: The lectures, coupled with discussions, present theoretical concepts, with a particular focus on applying spreadsheet programs and statistical software to solve economic problems using formulas, functions, and diagrams, as well as introducing fundamental statistical concepts.

Individual and group exercises: Through individual and group exercises, students apply what they have learned in practice, reinforcing their skills. Selection of the appropriate measurement level and corresponding measures of analysis, conducting samples and drawing conclusions from it, discussing the use and interpretation of statistics are deepened through these exercises.

Self-Study: In addition, Self-Study is introduced as a learning activity, providing students with

Session Plan:

EH	Date & Time	Room	Contact hours	Topics
EH1	23.02.2026 10:00 - 17:00	P.2.19	8	Intro: Syllabus, literature & software, recap empirical work, statistics & data science, creating numbers, scales, an analytical process; Using the logbook; gathering primary data (survey, sampling etc.)
EH2	24.02.2026 09:00 - 16:45	VC	9	Gathering primary data (survey, sampling etc.); gathering secondary data (repositories, validation, licences etc.); data handling & validation

25.02.2026				Descriptive Statistics I:
EH3 09:00 - 16:00	VC	8		Location and dispersion measures
26.02.2026				Descriptive Statistics II:
EH4 09:00 - 16:00	P.2.03	8		correlation measures
27.03.2026				
EH5 14:00 - 18:45	P.2.03	5		Buffer
18.04.2026				From statistics to decision making 1: Overview and examples
EH6 12:00 - 16:15	G.E.11	5		
16.05.2026				From statistics to decision making 2: Multicriteria evaluation approaches
EH7 09:00 - 16:00	G.2.12*	8		
22.05.2026				
EH8 14:00 - 18:15	VC	5		Q&A on assessments, buffer

1. attempt

Submission (Deliverable) (Prüfung)
written / - / 60,00%

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Submission (Deliverable) (Prüfung)
written / - / 40,00%

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2. attempt

Submission (Deliverable) (Prüfung)
written / - / 100,00%

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Self Study Time / Selbststudienzeit*

108 hours per semester / Stunden pro Semester

* The "workload" of a course comprises course attendance as

well as all work necessary to complete the course (course preparation), individual study, group work, research, assignments, exam preparation etc. (= Self-study time). The workload is expressed in ECTS (1 ECTS = 25 hours).

* Die "workload" für die positive Absolvierung einer Lehrveranstaltung umfasst die Präsenzzeiten sowie jene Zeiten, die für individuelle Vorbereitung, Gruppenarbeiten, Recherche, Verfassen von Arbeiten, Prüfungsvorbereitung etc. notwendig sind (= Selbststudienzeit). Die "workload" wird in ECTS ausgedrückt (1 ECTS = 25 Stunden).

Readings & media

Recommended Books:

Albright, S. C., & Winston, W. L. (2025). *Business analytics: Data analysis and decision making* (8th edn). Cengage.

Navarro, D., & Foxcroft, D. (2025). *Learning Statistics with jamovi: A Tutorial for Beginners in Statistical Analysis*. Open Book Publishers. <https://doi.org/10.11647/OBP.0333>

Sharpe, N. R., De Veaux, R. D., Velleman, P. F., & Berkowitz, J. (2019). *Business statistics: A first course* (Second Canadian edition). Pearson Canada Inc.

Also fun to read:

Huff, D. (1954). *How to Lie with Statistics*. Penguin UK.

Spiegelhalter, D. (2020). *The Art of Statistics: Learning from Data*. Pelican.
